Problem 1. Z & T, Ch 3, Pr. 6.6.

We consider 2 eq-ns & 2 dependent var-s, ρ & u. This problem is practice in showing that an obtained sol-n is, indeed, a sol-n by a direct subs-n. Show that

(1) $$\rho = \frac{g(x-ut)}{1+t f'(x-ut)}$$

is a sol-n to

(2) $$\rho_t + (\rho u)_x = 0$$

(3) $$\rho(x,0) = g(x)$$

where u is a sol-n to

(4) $$u_t + u u_x = 0$$

(5) $$u(x,0) = f(x) \quad \Rightarrow$$

(6) $$u = f(x-ut).$$

Sol-n:

First let me show that the sol-n in (1) satisfies the IC in (3):

(7) $$\text{IC: } \rho(x,0) = \frac{g(x-u\cdot 0)}{1+0\cdot f'(x-0\cdot t)} = \frac{g(x)}{1} = g(x) \checkmark.$$

Now to show that $\rho(x,t)$ in (1) satisfies the eq-n in (2), let's find ρ_t & $(\rho u)_x$:

$$\rho_t = \frac{\partial}{\partial t}\left[\frac{g(x-ut)}{1+tf'(x-ut)}\right] = \frac{g'(x-ut)\cdot\frac{\partial}{\partial t}(x-ut)\cdot\{1+tf'(x-ut)\} - g(x-ut)\cdot\frac{\partial}{\partial t}\{1+t\,f'(x-ut)\}}{\{1+tf'(x-ut)\}^2} =$$

$$= \frac{g'(x-ut)\cdot[0-(u_t\cdot t+u\cdot 1)]\cdot\{1+t\cdot f'(x-ut)\} - g(x-ut)\cdot\{0+[1\cdot f'(x-ut)+t\cdot f''(x-ut)\cdot(0-(u_t\cdot t+u\cdot 1))]\}}{\{1+t\cdot f'(x-ut)\}^2}$$

$$= \frac{[-u-t\cdot u_t]\cdot g'(x-ut)\cdot\{1+tf'(x-ut)\} - g(x-ut)\cdot\{f'(x-ut)-tf''(x-ut)\cdot[t\cdot u_t+u]\}}{\{1+t\cdot f'(x-ut)\}^2} =$$

$$= \frac{-u\cdot g'(x-ut) - t^2u_t\cdot g'(x-ut)\cdot f'(x-ut) - g(x-ut)\cdot f'(x-ut) + t\cdot g(x-ut)\cdot f''(x-ut)\cdot[tu_t+u]}{\{1+t\cdot f'(x-ut)\}^2} =$$

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$$= \frac{-u\cdot g'(x-ut) - t^2u_t\cdot g'(x-ut)\cdot f'(x-ut) - g(x-ut)\cdot f'(x-ut) + t^2u_t\cdot g(x-ut)\cdot f''(x-ut) + t\cdot u\,g(x-ut)\cdot f''(x-ut)}{\{1+t\cdot f'(x-ut)\}^2 = \text{denom.}}$$

Now for $(\rho u)_x$, note that $(\rho u)_x = \rho_x u + \rho u_x$ using the product rule.

$$(\rho u)_x = \rho_x u + \rho u_x = \frac{\partial}{\partial x}\left[\frac{g(x-ut)}{1+tf'(x-ut)}\right]\cdot u + \frac{g(x-ut)}{1+tf'(x-ut)}\cdot\frac{d}{dx}\left[f(x-ut)\right] =$$

$$= u\cdot\frac{g'(x-ut)\cdot\frac{\partial}{\partial x}[x-ut]\cdot\{1+tf'(x-ut)\} - g(x-ut)\cdot\frac{\partial}{\partial x}[1+tf'(x-ut)]}{\{1+tf'(x-ut)\}^2} + \frac{g(x-ut)}{1+tf'(x-ut)}\cdot f'(x-ut)\cdot\frac{\partial}{\partial x}[x-ut]$$

$$= u\cdot\frac{g'(x-ut)\cdot[1-t\cdot u_x]\cdot\{1+tf'(x-ut)\} - g(x-ut)\cdot[t\cdot f''(x-ut)\cdot(1-tu_x)]}{\{1+t\cdot f'(x-ut)\}^2} + \frac{g(x-ut)\cdot f'(x-ut)\cdot[1-tu_x]}{1+tf'(x-ut)}$$

$$\overset{B}{=} \frac{[u\cdot g'(x-ut) - t\cdot u\cdot u_x\, g'(x-ut)]\cdot\{1+tf'(x-ut)\} - g(x-ut)[t+f''(x-ut)\cdot(1-tu_x)] + g(\cdot)\cdot f'(\cdot)\cdot[1-tu_x]\cdot\{1+t\cdot f'(\cdot)\}}{\{1+tf'(x-ut)\}^2 = \text{denom.}}$$

So, pay attention to the results A & B for ρ_t & $(\rho u)_x$ respectively. Also, note that A & B share the common denominator. So only consider the numerators of A & B:

$$\rho_t + (\rho u)_x = \frac{1}{\text{denom}}\Big\{\Big[-\underline{u g'(x-ut)} - t^2 u_t\, g'(x-ut)\cdot f'(x-ut) - g(x-ut)\cdot f'(x-ut) + t^2 u_t\, g(x-ut)\cdot f''(x-ut) + t\cdot u\, g(x-ut)\cdot f''(x-ut)\Big] +$$

$$+\Big[\underline{[u\cdot g'(x-ut)}] - tu\cdot u_x\, g'(x-ut)]\cdot\{1+tf'(x-ut)\} - g(x-ut)\cdot[t+f''(x-ut)\cdot(1-tu_x)] + g(x-ut)\cdot f'(x-ut)\cdot[1-tu_x]\cdot\{1+t\cdot f'(x-ut)\}\Big]\Big\} =$$

Now let me substitute $\boxed{u_t = -u u_x}$ derived by (4) into the eq-n:

$$= \frac{1}{\text{denom}} \cdot \Big\{ \Big[-t^2(-u u_x) \cancel{g'(x-ut)} \cdot f'(x-ut) - g(x-ut) \cdot f'(x-ut) + t^2(-u u_x)\, g(x-ut) \cdot f''(x-ut) + t \cdot u \cdot g(x-ut) \cdot f''(x-ut)$$

$$t \cdot u u_x \cdot g'(x-ut) - \underline{t^2 u u_x \cancel{g'(x-ut)} \cdot f'(x-ut)} + t \cdot u g'(\cdot) f'(\cdot) - t \cdot g(\cdot) - g(\cdot) \cdot f''(\cdot) - t u_x g(\cdot) \cdot f''(\cdot) + \big[g(\cdot) \cdot f'(\cdot) - t u_x g(\cdot) f'(\cdot) \big] \cdot \{1 + t f'\} \Big] \Big\}$$

$$= \frac{1}{\text{denom}} \Big\{ \Big[\boxed{g \cdot f'} - t^2 u u_x g \cdot f'' + t u g f'' \Big] - t u u_x g' + t u g' f' - t g - g \cdot f'' - t u_x g f'' + \underline{[g f' - t u_x g f']} \{1 + t f'\} \Big\}$$

$$= \ldots = \frac{1}{\text{denom}} \cdot \{0\} = 0$$

I know these terms all should cancel eventually into <u>zero</u> provided ρ & u are solution to the corresponding problems. I've double checked with Professor, all derivatives are true but matter of calculation. I hope you understand the process if very time consuming, so decided to stop here. Actually, I'm trying on it since Monday.

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